

GeM-CompliAI Platform Operational Guide

Comprehensive Functional Breakdown: Buyer Desk vs. Seller Desk Authority & Capabilities

Target Governance: GFR 2017 & GeM GTC v4.0

Authentication: NIC SSO / Class-3 DSC / Aadhaar e-KYC

Platform Version: 2.0 Real-Time Engine

1. Role-Based Access Control (RBAC) Architecture

The **GeM-CompliAI** platform strictly segregates procurement powers between **Buyer Authorities** (Procurement Officers, Indenting Officers & CPSE Buyers) and **Seller Entities** (Enterprises, MSEs, Startups, and OEMs). This architecture prevents conflicts of interest and guarantees full compliance with the General Financial Rules (GFR) 2017, Public Procurement Policy for MSEs, and Make in India (MII) mandates.

2. Buyer Desk Capabilities (Procuring Officers & Indent Creators)

The Buyer Desk provides government officials with end-to-end statutory verification, automated AI rule extraction, contract administration, and binding qualification tools.

- **A. Create & Publish AI-Parsed Tenders (RFPs):** Upload custom tender specifications (PDF/Text). The AI Engine automatically extracts turnover thresholds, EMD exemptions, Make in India local content percentages, and 14 statutory verification checklist clauses with end-to-end AES-256 encryption.
- **B. Multi-Portal Statutory Cross-Reconciliation:** Cross-verify bidder credentials in real-time across 8 official databases: GSTN return filings (GSTR-3B/1), Ministry of MSME Udyam classification, Income Tax PAN/ITR-6, MCA21 CIN registry, EPFO/ESIC compliance, and Central CVC/GeM Debarment registry.
- **C. AI Compliance Scoring & Tamper Forensics:** Inspect OCR entity extractions, forensic font consistency scores, and SHA-256 cryptographic document fingerprints to catch fabricated certificates (e.g. counterfeit CA UDIN numbers or forged OEM authorizations).
- **D. Record Legally Binding Officer Decisions:** Officially QUALIFY or DISQUALIFY participating bidders with mandatory statutory reasoning and digitally sign evaluations using Class-3 Digital Signature Certificates (DSC) anchored in the tamper-evident Audit Vault.
- **E. Direct Purchase & L1 Marketplace Orders:** Procure items under GFR Rule 149 (Direct Purchase up to **₹5 Lakhs** / automated L1 comparison) with automated delivery milestone tracking and contractual terms.
- **F. Issue Consignee Receipt and Acceptance Certificates (CRAC):** Issue digital CRAC upon physical site inspection. Issuing CRAC opens the 10-day automated PFMS payment disbursement mandate under GeM GTC.
- **G. Incident Management System (IMS) & SCN Issuance:** Raise statutory Show-Cause Notices (SCN) for delivery delays, defective items, or fraudulent documents, initiate seller remediation windows, and recommend central debarment.

3. Seller Desk Capabilities (Enterprises, MSEs, Startups & OEMs)

The Seller Desk empowers suppliers to participate in national tenders, monitor real-time Reverse Auctions, run pre-submission compliance self-checks, and manage invoicing.

- **A. Bid Submission with Automated Document AI:** Submit e-Bids against active central tenders. Upload 9 mandatory statutory documents (PAN, GST REG-06, Udyam, CA Audited Turnover with UDIN, ITR-6, MII Local Content declaration, OEM Authorization, and EPFO challans).
- **B. Real-Time Reverse Auction (RA) Room Bidding:** Participate in live competitive RA rooms with real-time countdown clocks, minimum decrement rule validation, dynamic L-1 ranking shifts, and instant outbid alerts.
- **C. Pre-Check Compliance Self-Audit:** Run the AI verification engine prior to tender submission to identify document discrepancies, missing turnover criteria, or invalid UDIN formatting before formal evaluation.

- **D. Instant Compliance Certificate Generation:** Download cryptographically sealed and QR-verifiable GeM Bid Compliance Certificates confirming verified MSE classification and zero-debarment status.
- **E. Purchase Order Fulfillment & Logistics Tracking:** Track Direct Purchase contracts, view consignment status, monitor consignee inspection progress, and receive CRAC approval notifications.
- **F. GST Tax e-Invoice Generation:** Generate official GST-compliant tax invoices with unique IRN numbers upon CRAC acceptance for automated settlement via Public Financial Management System (PFMS).
- **G. Incident Response & Show-Cause Notice Clarifications:** Submit formal representations and upload supporting evidence in response to active IMS complaints within the 7-day statutory clock.

4. Buyer Desk vs. Seller Desk Authority Comparison Matrix

Comprehensive comparison of functional permissions across the entire procurement lifecycle:

Publish New Tender / RFP	FULL ACCESS Create RFPs & AI checklist	RESTRICTED View & participate only	GFR Rule 144 / 161
Statutory Portal Recon (8 APIs)	FULL ACCESS Inspect live API snapshots	SELF-VIEW ONLY View own verified badges	CVC / GSTN / MSME
Submit e-Bid Packages	RESTRICTED Evaluation role only	FULL ACCESS Upload 9 statutory forms	GeM GTC Cl. 4
Reverse Auction (RA) Bidding	MONITOR ONLY Observe live decrement feed	ACTIVE BIDDER Place decrement e-bids	GFR Rule 153 / RA SOP
Document AI Tamper Inspector	FULL ACCESS Forensic font & SHA-256	PRE-CHECK TOOL Self-audit before submission	STQC / ISO 27001
Qualification Decisions (DSC)	BINDING AUTHORITY Qualify/Disqualify with DSC	RESTRICTED Cannot self-qualify	IT Act 2000 Sec 3A
Direct Purchase Execution	PURCHASER Award POs up to ₹5 Lakhs	ORDER FULFILLMENT Supply goods & logistics	GFR Rule 149(i)
Issue Consignee CRAC	CONSIGNEE ONLY Inspect & issue CRAC	RESTRICTED Receives payment trigger	GeM GTC Cl. 12
Generate GST e-Invoice	VERIFY & APPROVE Approve for PFMS release	INVOICE CREATOR Generate IRN-stamped bill	GSTN E-Invoicing
Incident Management (IMS)	RAISE SCN NOTICE Demand explanation	SUBMIT RESPONSE Submit statutory defense	GeM Incident Policy
Audit Vault & Hash Logs	AUDITOR ACCESS Full cryptographic chain	RESTRICTED Confidential auditor ledger	CAG Guidelines

5. Cryptographic Security & Statutory Governance

All interactions across both desks are protected by multi-layered national security standards:

- **Class-3 DSC Digital Signatures:** Every procurement decision and contract award is sealed with SHA-256 signatures.
- **Aadhaar e-KYC & Mobile 2FA:** Zero-trust identity authentication for all verified procurement officers.
- **Tamper-Evident Audit Vault:** Immutable event logging for CAG and Vigilance inspection.
- **Automated PFMS Settlement:** 100% timely payment compliance tied strictly to digital CRAC issuance.